

The Half Angle Identities and More Double Angle Problems (Homework)

① If θ is in standard position, and it terminates in QIV ($270^\circ < \theta < 360^\circ$), find the value of each expression below if $\cos \theta = \frac{1}{4}$.

i) $\cos\left(\frac{\theta}{2}\right)$

ii) $\sin\left(\frac{\theta}{2}\right)$

iii) $\tan\left(\frac{\theta}{2}\right)$

② If β is in standard position, and it terminates in QII ($90^\circ < \beta < 180^\circ$), find the value of each expression below if $\cos \beta = -\frac{1}{5}$.

i) $\sin\left(\frac{\beta}{2}\right)$

ii) $\cos\left(\frac{\beta}{2}\right)$

iii) $\csc\left(\frac{\beta}{2}\right)$

③ If β is in standard position, and it terminates in QIII ($180^\circ < \beta < 270^\circ$), find the value of each expression below if $\sin \beta = -\frac{\sqrt{11}}{6}$.

i) $\sin\left(\frac{\beta}{2}\right)$

ii) $\cos\left(\frac{\beta}{2}\right)$

④ If B is in standard position, and it terminates in QI ($0^\circ < B < 90^\circ$), find the value of each expression below if $\sin B = \frac{2}{3}$.

i) $\cos(2B)$

ii) Find the value of $\sin(2B)$ using the double angle formula below.

$$\sin(2B) = 2\sin B \cos B$$

iii) Find the value of $\sin(2B)$ using the value of $\cos(2B)$ that you found in part one of this problem.

⑤ If W is in standard position, and it terminates in QIII ($180^\circ < W < 270^\circ$), find the value of each expression below if $\cos W = -\frac{\sqrt{6}}{4}$.

i) $\cos(2W)$

ii) Find the value of $\sin(2W)$ using the double angle formula below.

$$\sin(2W) = 2\sin W \cos W$$

iii) Find the value of $\sin(2W)$ using the value of $\cos(2W)$ that you found in part one of this problem.

Homework Answers

$$\textcircled{1} \text{ i) } \boxed{-\frac{\sqrt{10}}{4}}$$

$$\text{ii) } \boxed{\frac{\sqrt{6}}{4}}$$

$$\text{iii) } \boxed{-\frac{\sqrt{15}}{5}}$$

$$\textcircled{2} \text{ i) } \boxed{\frac{\sqrt{15}}{5}}$$

$$\text{ii) } \boxed{\frac{\sqrt{10}}{5}}$$

$$\text{iii) } \boxed{\frac{\sqrt{15}}{3}}$$

$$\textcircled{3} \text{ i) } \boxed{\frac{\sqrt{33}}{6}}$$

$$\text{ii) } \boxed{-\frac{\sqrt{3}}{6}}$$

$$\textcircled{4} \text{ i) } \boxed{\frac{1}{9}}$$

$$\text{ii) } \boxed{\frac{4\sqrt{5}}{9}}$$

$$\text{iii) } \boxed{\frac{4\sqrt{5}}{9}}$$

$$\textcircled{5} \text{ i) } \boxed{-\frac{1}{4}}$$

$$\text{ii) } \boxed{\frac{\sqrt{15}}{4}}$$

$$\text{iii) } \boxed{\frac{\sqrt{15}}{4}}$$